Mechanical Engineering
Carnegie Mellon University

Senior Capstone Design Project

Automated Resistor Sorter

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MOTIVATION

After every electronics project comes the same problem: a **pile of unsorted resistors**.

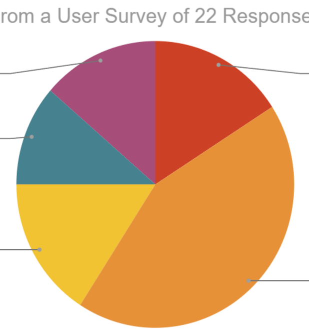
- Cleaning and sorting resistors after projects is **time-consuming**.
- Resistors are cheap; users more often decide to discard used. resistors and buy new ones between projects, resulting in **waste**.
- Existing projects are limited by slow sorting throughput and require manual, one-by-one resistor input.

CUSTOMER NEEDS & SPECS

Goal: Create an automated sorting system that enables users to **accurately sort batches of resistors** with **minimal effort**.

Most Time-Consuming Steps

From a User Survey of 22 Responses

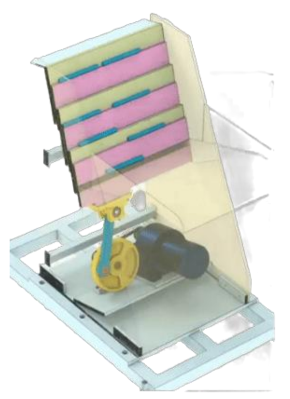


- #1 Need: **Accuracy**
- #2 Need: **Large Storage**
- #3 Need: **Ease of Use**

CONCEPTS & SELECTION

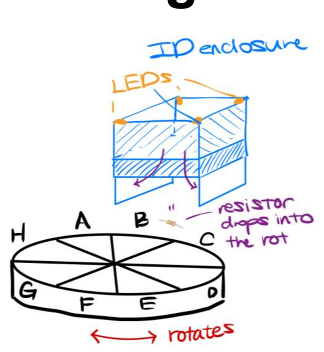
Singulation

Vibratory Spiral Ramp
- Difficult singulation due to overlap and clusters
- Slow and unpredictable feed rate

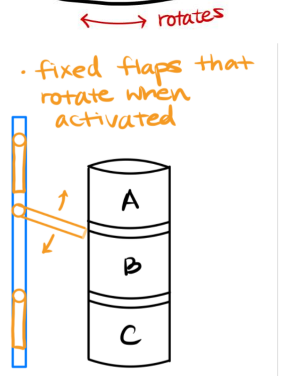


Slider-Crank Step Ladder
- Precise indexing ensures single resistor per step
- Positive mechanical motion reduces jamming risk

Storage

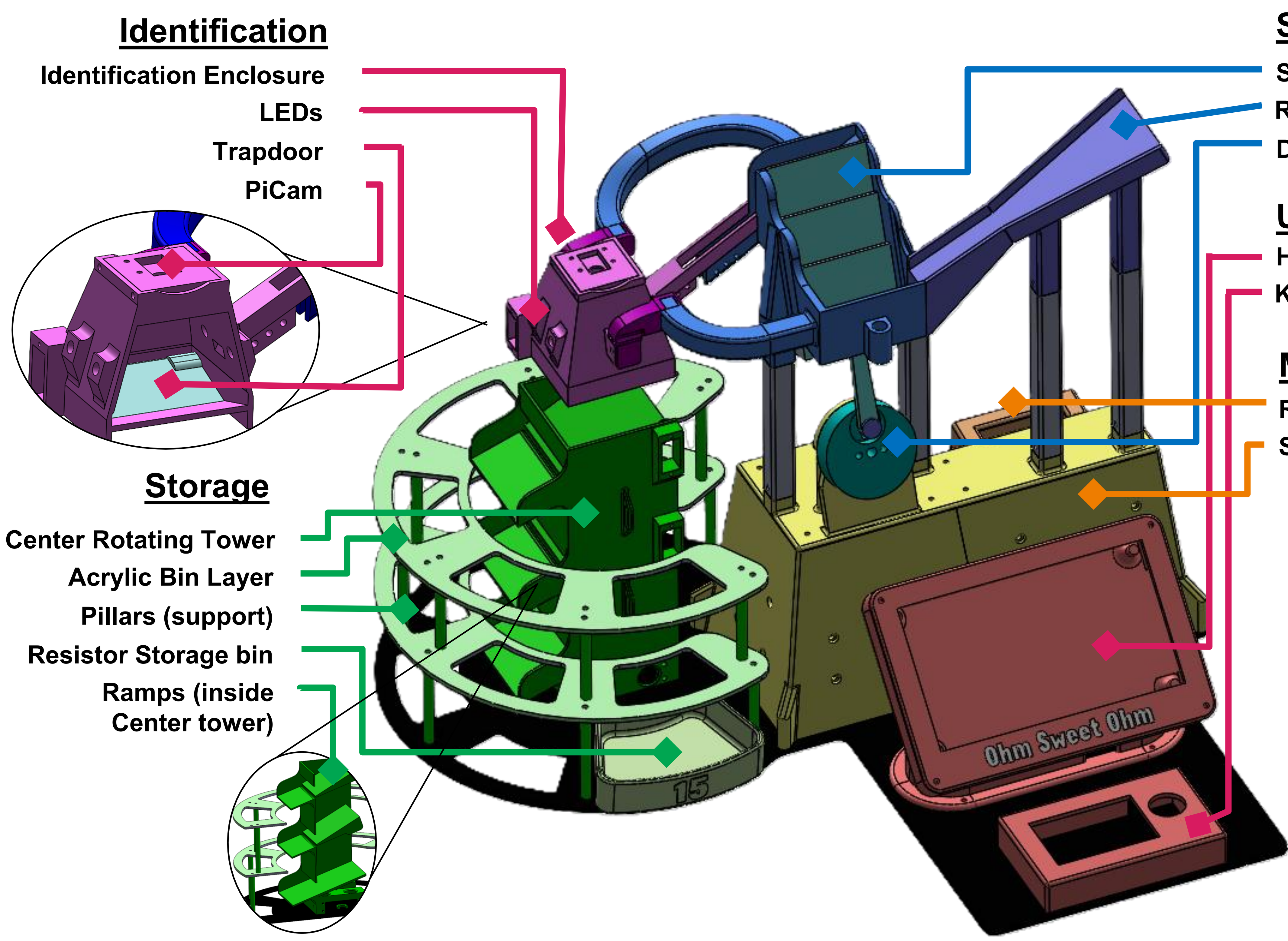


Rotating Bins around Ramp
- Limited storage capacity
- Slower



Rotating Central Tower:
- Larger + scalable storage capacity
- Efficient turning capabilities

HARDWARE INTERFACE & SPECS



Identification
- Identification Enclosure
- LEDs
- Trapdoor
- PiCam

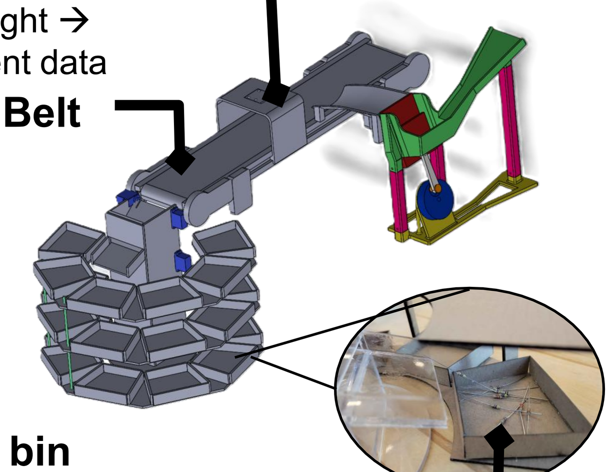
Storage
- Center Rotating Tower
- Acrylic Bin Layer
- Pillars (support)
- Resistor Storage bin
- Ramps (inside Center tower)

Singulation
- Step Ladder
- Ramp (accepts resistors in bulk)
- DC Motor (12V 0.6A 30RPM)

UI Interface
- HDMI Display
- Keypad (9 inputs)

Miscellaneous
- Raspberry Pi 4 Model B
- Standoff

Iteration 1

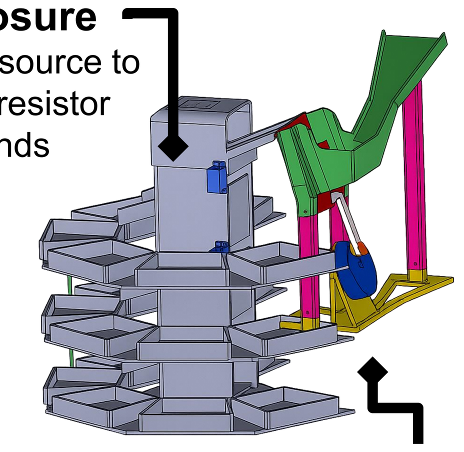


CV enclosure
- Variable light → inconsistent data

Conveyor Belt
- Too large

Storage bin
- Corners caused resistors to have inconsistent orientation
- Led to jams and reduced resistor capacity

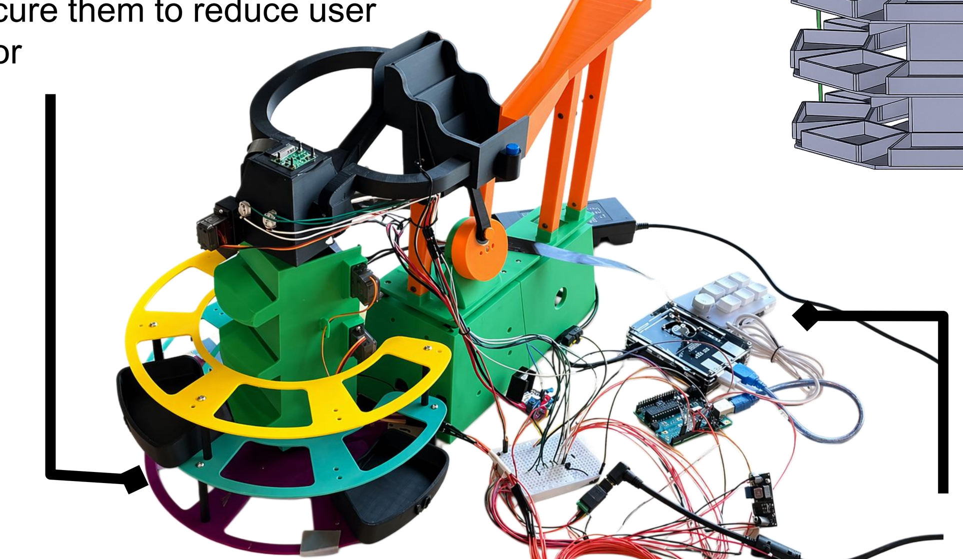
Iteration 2



CV enclosure
- No light source to capture resistor color bands

Standoff Needed
- Height difference between the systems

Iteration 3



Inconsistent Setup
- Different subsystem orientations can cause errors
- Secure them to reduce user error

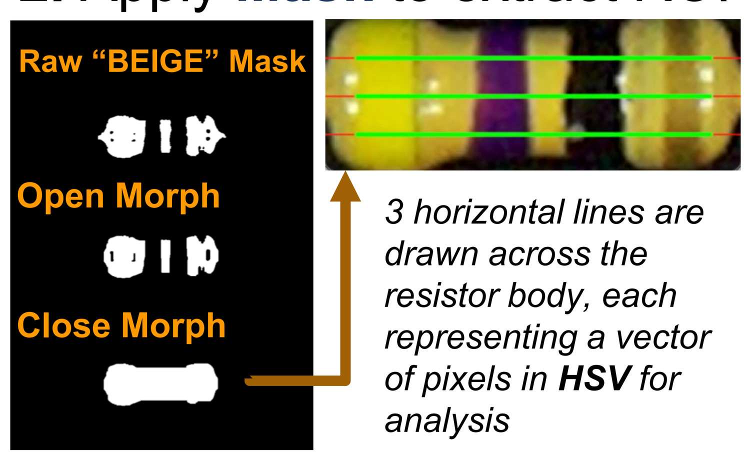
Loose Electronics
- Reduce ease of use and complicate setup

Dangling Wires
- Affects accuracy of the storage subsystem

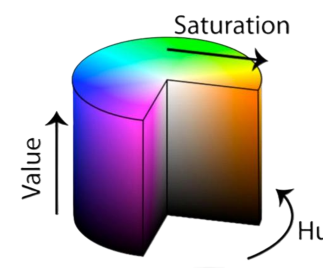
COMPUTER VISION

1. Capture frame with PiCam

2. Apply **Mask** to extract ROI

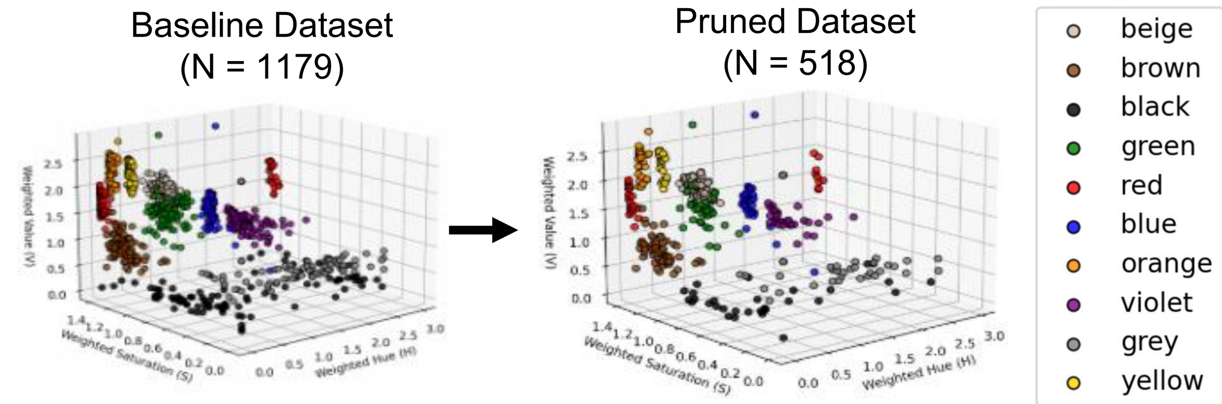


3 horizontal lines are drawn across the resistor body, each representing a vector of pixels in HSV for analysis



Iterating through every other pixel of the pixel vectors to reduce inference time by half

3. **KNN** to predict color for each pixel



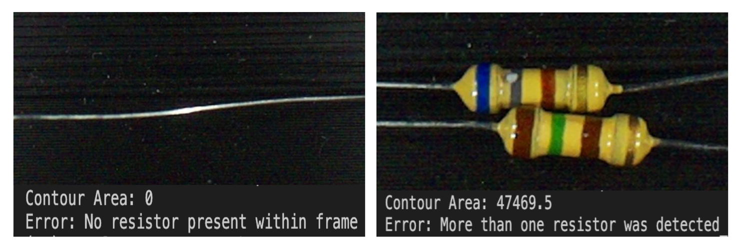
KNN **computation time** is strictly proportional to **dataset size**
- 50% dataset size → 50% computation time (3.66 sec → 1.80 sec)

4. **Vote** on a single 3-color sequence & calculate resistance

```
--- Running Horizontal Multi-Line Analysis ---
Sequence 1: ('yellow', 'violet', 'black')
Sequence 2: ('yellow', 'violet', 'black')
Sequence 3: ('yellow', 'violet', 'black')
Final Sequence: ['yellow', 'violet', 'black']
Resistance:  47 Ohms
```

Additional Features

1. Edge Case Handling



No resistor | Double Resistor

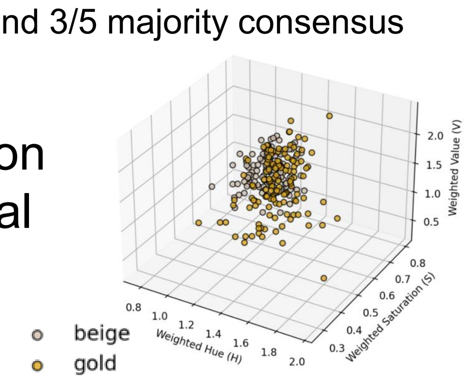
2. Deep Scan

```
--- Running Horizontal Multi-Line Analysis ---
Sequence 1 (Y-coord: 48): ('red', 'black', 'brown')
Sequence 2 (Y-coord: 24): ('brown', 'red', 'black', 'brown')
Sequence 3 (Y-coord: 72): ('brown', 'red', 'black', 'brown')

Ambiguous Read!!! Triggering Deep Scan...
Sequence 4 (Y-coord: 36): ('red', 'black', 'brown')
Sequence 5 (Y-coord: 60): ('red', 'black', 'brown')
Phase 2 Pass! ('red', 'black', 'brown') achieved 3/5 consensus
```

Resolves inconclusive reads via 5-line sampling and 3/5 majority consensus

3. Gold Detection Removal



Allows resistance classification more robust because it resolves beige/gold similarity

RESULTS & CONCLUSIONS

- Fully **automated bulk resistor sorting** achieved
- Highly **reliable** correct sorting from start to finish
- **Scalable** storage achieved

Subsystem	Accuracy			Throughput
Singulation	0 output	1 output	2 output	~ 2 seconds
	4.7%	94.2%	.01%	
Identification	Classification	Perfect 3/3		1.80 sec
	97.6%	83.3%		
Storage	In bin	Out of Bin		~2 sec
	97.5%	2.5%		

FUTURE WORK

- Develop a custom **PCB** for better reliability and organization.
- Redesign for greater **compactness** and **safety**.
- Utilize a more powerful **GPU** to improve processing speed and performance.
- Generalize system to accept **more types of resistors**.